

Words Speak Louder Than Order: A Behavioral Evaluation of Gemma 4

A Fully Counterbalanced Targeted Behavioral Study of Gemma 4-e4b

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Abstract

When a language model receives two conflicting documents as input, how does it decide which one to prioritize? Does it rely on how the sources are framed or the presentation order of the documents? We evaluated this behavior on **Google’s pre-trained Gemma 4-e4b** model across a targeted behavioral suite ($n = 13$ items, 784 forward passes in short, single-turn contexts) using a completely counterbalanced experimental design. This setup allowed us to mathematically isolate the specific effects of source framing and reading position, while ensuring the model’s natural vocabulary biases were canceled out.

Across ten test conditions, we discovered the following:

1. **Source framing heavily overpowers reading position.** When directly competing, the semantic framing of a source (such as presenting it as an official guideline or a fresh update) had a significantly stronger impact on the model’s final answer than the presentation order of the document.
2. **The model favors the first document it reads, but this bias is highly variable.** While the model consistently demonstrated a *primacy effect* (preferring the first document presented), the actual strength of this bias fluctuated by at least a factor of 5 based solely on the surface wording.
3. **Overall structural repetition, not short copy-cues, drives positional bias.** The model’s preference for the first document is not a mechanical reaction to short, repetitive trigger phrases, such as “is [Answer]”. However, the primacy effect does increase significantly when the two competing documents are structurally identical, using word-for-word verbatim templates. Introducing variation in the overall wording between the two sources reduces this positional bias.

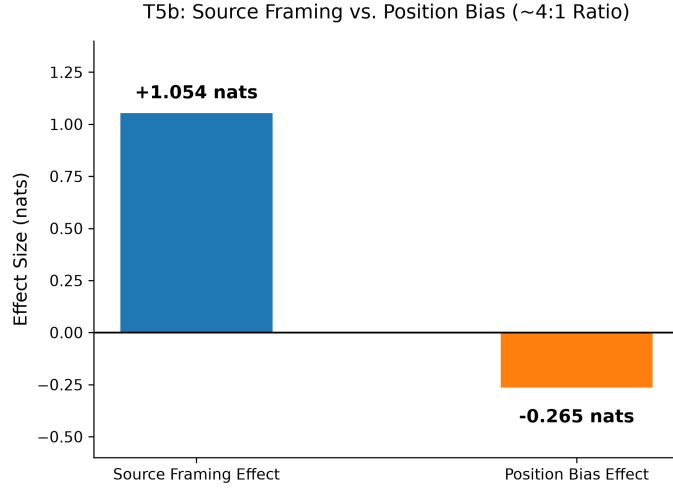


Figure 1: Comparison of source framing and positional bias in Gemma 4. When competing directly, the magnitude of the semantic framing effect outweighed the presentation order bias by a statistically significant 4-to-1 margin.

Code, raw data logs, and prompt templates for all experiments are publicly available on GitHub (gemma-priority-probing). Finally, we note that all of our results are strictly behavioral observations about text processing.

1. Introduction

Language models are often forced to read conflicting information at the same time, such as an outdated official document right next to a fresh, informal team update. Understanding which source the model ultimately prioritizes is critical for building reliable tools, but accurate measurement remains a challenge due to overlapping biases (Zheng et al., 2023; Wang et al., 2023; Sclar et al., 2024).

Typically, three distinct factors get mixed up and masquerade as the model’s source prioritization:

1. The model’s natural preference for certain words over others (Sclar et al., 2024).
2. The order in which the information is read by the model (Liu et al., 2023; Zheng et al., 2023).
3. The authoritative or recent tone used in the text (Wan et al., 2024).

The stakes for resolving this ambiguity extend far beyond standard benchmarking. In production Retrieval-Augmented Generation (RAG) systems, failing to

decouple these variables can lead to critical failures, such as an outdated legacy document silently overriding a crucial but informally written update simply because it sounds more official. Furthermore, this behavior introduces significant security vulnerabilities: If a model defers primarily to semantic authority, malicious actors can easily hijack its output by planting deceptive instructions under authoritative headers. If these confounding factors are not carefully separated, evaluators risk misinterpreting a basic positional artifact or a vocabulary preference as a genuine semantic choice.

Standard evaluations attempt to mitigate these overlapping biases by simply swapping the presentation order of the documents (Zheng et al., 2023). **However, this mitigation is insufficient.** Simple order-swapping leaves a model’s inherent vocabulary preferences completely entangled with its semantic choices, creating massive artificial artifacts in the resulting data. To solve this measurement problem, we designed a fully counterbalanced experiment to mathematically isolate these variables from the ground up. By running every single test item through a four-way crossing matrix, we separated these influences by design, rather than relying on statistical models to estimate the impact of biased data after it was collected. Our evaluation focused on short, single-turn contexts across 10 conditions ($n = 13$ items, 784 forward passes).

Related work:

- Position bias in document ordering is well characterized (Liu et al., 2023; Zheng et al., 2023), as is the sensitivity of LLM evaluations to spurious features and meaning-preserving prompt variation (Sclar et al., 2024). Our $\sim 5\times$ spread in position-bias magnitude across templates extends Sclar’s finding from *accuracy* to *bias magnitude*.
- Wan et al. (2024) report that models largely ignore stylistic authority cues; we find an *explicit* role label outweighs position by a roughly 4:1 margin in our isolated test condition, suggesting label-level and style-level authority signals dissociate.
- We adopt TOST equivalence testing (Lakens, 2017), standard in psychology but rare in LLM evaluation.

2. Methods

This section outlines our complete experimental framework. It covers the formulas used to pull apart and isolate different model biases, the specific variables that we adjusted to create our ten test conditions, and the rigorous testing setup used to ensure our data was clean, reliable, and reproducible.

2.1 Contrast algebra

Every test showed the model two text blocks. One block was labeled **Recent** (the newer update) and one was labeled **Stale** (the older one). Each block explicitly recommended one of the two words. After both blocks, we always asked the exact same question, and we forced the model to choose between two single words (for example, **Agile** and **Waterfall**).

To illustrate, a single test prompt might conceptually look like this:

Official Guideline (Stale): Use Agile.
 Team Update (Recent): Use Waterfall.
 Question: Which should we use?

(For the exact wording variations and the full four-way crossing matrix, see Appendix C and Appendix D).

We then measured e , the model’s preference for **Waterfall** over **Agile**, in log-probability units. Positive e meant it leaned **Waterfall**. We ran each word pair through four versions of the same test. Two things were swapped: which block to print first, and which word to carry the **Recent** label.

Table 1: The four-way counterbalanced experimental design.

Cell	First block	Second block
e_1	Stale (Agile)	Recent (Waterfall)
e_2	Recent (Waterfall)	Stale (Agile)
e_3	Stale (Waterfall)	Recent (Agile)
e_4	Recent (Agile)	Stale (Waterfall)

Because all four combinations appear once, we can pull apart four separate influences by calculating their average across the grid:

- **Average base word preference** $= (e_1 + e_2 + e_3 + e_4)/4$: Does the model simply like the word **Waterfall** more than **Agile**, no matter the context?
- **Average source framing** $= (e_1 + e_2 - e_3 - e_4)/4$: How much does being labeled **Recent** help a word win?
- **Average serial position** $= (e_1 - e_2 - e_3 + e_4)/4$: Does the model favor whichever word it read last (positive) or first (negative)?

- **Residual check (Block Order (Null Control))** = $(e_1 - e_2 + e_3 - e_4)/4$:
A sanity test. Nothing real should live here, so it should come out near zero. If it doesn't, something in the design leaked.

Language models carry strong built-in word preferences. If the model has an inherent preference for the word **Waterfall**, that baseline bias shows up in all four cells equally. Because our experimental design fully counterbalances which word acts as the **Recent** update, this vocabulary bias cancels algebraically within the framing and serial formulas. (For a demonstration of the massive artifact that occurs without this counterbalancing, see Appendix A). This leaves only the targeted behavioral effects we aim to measure.

Note: these formulas give the half-effect (the pull away from a neutral midpoint). The full gap between the two extremes is 2x the number shown. Every table in this paper reports half-effects.

2.2 The ten test conditions

Each test condition is built by setting three dials. Every condition uses the same underlying disagreement (two blocks that recommend different words), so any difference between conditions comes from *presentation*, not content.

Table 2: Experimental variables used to construct the test conditions.

Dial type	Settings	What it changes
Header wording	<i>neutral</i> , <i>framed</i>	Whether the blocks are: neutral: labeled with an unbiased identifier blandly (e.g., “Document A”) framed: labeled as an official guideline (authority) or a fresh team update (recency). (e.g., authority: “Official Guideline:”) (e.g., recency: “Team Update:”)
Body wording	<i>verbatim</i> , <i>mixed</i> , <i>loaded</i>	The two block bodies are: verbatim: word-for-word identical except the answer word (e.g., “The required system is [Answer].”) mixed: reworded using difference sentence structures for each block. (e.g., Block 1: “The required system is [Answer],”) (e.g., Block 2: “[Answer] is the recommended method.”) loaded: carries the framed meaning in the sentence itself. (e.g., “Official guidelines state the required system is [Answer].”)
Cue count	0, 1, 2	How many times the exact word pattern appears. (e.g. “is {e}”)

By combining the settings above, we created the following 10 qualifying test conditions:

Table 3: Configurations of the 10 qualifying test conditions.

Test Condition	Header Wording	Body Wording	Cue Count
T1	neutral	verbatim	2
T1c	neutral	verbatim	0
T2	neutral	mixed	1
T2c	neutral	mixed	0
T3	framed	verbatim	2
T3c	framed	verbatim	0
T4	framed	mixed	1
T4c	framed	mixed	0
T5	framed	loaded	2
T5b	framed	loaded	0

(Note: For detailed definitions of these specific test conditions and how they are evaluated in our results, please see Appendix E.)

Details:

- The code counts cues at runtime and halts if the number of cues found does not match the expected target for that condition (0, 1, 2). For our tests, all conditions passed.
- The header establishes the context, while the body remains neutral in T1 through T4c.
- The experiment fully counterbalanced which specific answer word played which role.
- In role-bound conditions (T5 & T5b), the body actively drives the bias. Because the body text contains the authoritative or recent tone (e.g., “Official guidelines state that...”), it is role-bound and permanently tied to its corresponding header. We deliberately did not counterbalance (swap) these bodies between roles, because doing so would average the manipulation to zero.
- Of the 18 theoretical combinations of these settings, 8 were excluded because they are logically contradictory or structurally impossible. For example, a neutral header cannot be paired with a loaded body (because the body itself carries the authoritative or recent manipulation), and verbatim text cannot contain exactly 1 cue (since the two text blocks must remain word-for-word identical, meaning cues must be symmetrical).

2.3 The testing setup

To overcome the confounding variables present in standard evaluations (discussed in Section 1), this section details the custom experimental framework and environment used to evaluate the model’s behavior. We defined the candidate items tested and outlined the strict inclusion rules designed to filter out the model’s natural vocabulary preferences and other experimental artifacts. Furthermore, we established the statistical procedures utilized to calculate our confidence intervals and test for equivalence, and we provided the complete system specifications required to fully replicate the study.

2.3.1 Test items We curated a targeted pool of 15 candidate English project-management noun pairs (e.g., Scrum / Kanban, Agile / Waterfall) rather than sampling from vocabulary. We used the hand-curation approach to help ensure pairwise semantic symmetry, with the intent to guarantee that either word served as a plausible, interchangeable competing answer to the identical question across all four counterbalanced configurations. Additionally, we wanted to minimize baseline prior differentials to reduce estimation variance. Two candidate pairs were subsequently excluded by programmatic validation filters (see rules below).

1. Scrum / Kanban
2. Agile / Waterfall
3. Roadmap / Milestone
4. ~~Spec / Brief~~
5. Scope / Scale
6. Phase / Stage
7. Cycle / Sprint
8. Board / Grid
9. Matrix / Table
10. Guide / Manual
11. Index / Log
12. Deck / Pitch
13. ~~Queue / Roster~~
14. Plan / Draft
15. Chart / Graph

2.3.2 Rules We applied these rules to our tests:

1. Every answer must be a single token so that the probability we read off is the probability of the whole word and nothing else. ~~Queue/Roster~~ failed (one token vs. two) and was dropped.
2. To ensure consistency, an item was kept only if it passed the checks in all ten conditions. This means every table in the paper analyzes the exact same dataset. The ~~Spec/Brief~~ pair (T5b, Order = 0.602) failed this

requirement in one condition and was excluded.

3. We flagged any word pair that failed one of two sanity checks in a given condition:
 1. The model naturally heavily preferred one word over the other ($|\text{Token}| > 2.0$).
 2. Our control test detected an unexpected bias leak ($|\text{Order}| > 0.50$).
4. We re-ran every comparison (contrast) on the resulting unfiltered set as a pre-planned robustness check.

2.3.3 Confidence intervals To build confidence intervals, we used **BCa bootstrap intervals, 9,999 resamples, seed 42**. In plain terms:

1. Take test results calculated for each of the 13 word pairs used in the experiment.
2. Randomly pick 13 word pairs, putting each pair back into the pool before drawing the next, and calculate their average.
3. Do that 9,999 times to see how the results spread out.
4. The central 95% of this distribution defines our confidence interval, which is adjusted for skewness using the standard BCa correction. To guarantee exact reproducibility for future replications, we initialized the process with a fixed random seed.

If a confidence interval did not include zero, it indicated that the direction of the effect remained consistent across all plausible resampled variations of our test items.

While a confidence interval can confirm whether an effect exists, it does not answer whether the effect is small enough to be considered trivial. To answer that, we used a statistical method called TOST (two one-sided tests), setting a triviality bound of ± 0.15 nats. The test only confirms the effect is trivial if we are statistically confident ($p < 0.05$) that the entire 90% interval fits inside that narrow ± 0.15 range. That test returned **equivalent=False in all ten conditions**. This means that we never demonstrated that any effect is trivially small, but it also does *not* mean we proved the effects are large.

Instead, the observed bias is directionally negative: across 8 of the 10 conditions, the 90% confidence interval falls entirely below -0.15 nats. The only exceptions are T4 (90% CI $[-0.262, -0.082]$) and T5 ($[-0.293, -0.091]$).

*Reminder on units: a **nat** measures how hard the model leans. $+1$ nat ≈ 2.72 -to-1 odds; $+2$ nats ≈ 7.4 -to-1. All tables report **half-effects**, so the full swing between the two extremes is twice the number shown.*

2.3.4 Environment

- macOS-15.7.7 (arm64)
- Python 3.12.13
- torch 2.13.0
- transformers 5.14.1
- numpy 2.5.1
- scipy 1.18.0
- Full 32-bit precision, evaluation mode, no random sampling.
- Run under the **venv_torch** environment (not `./venv`) for J-Lens toolkit compatibility.
- **784 forward passes** total.
- Run of record: **2026-07-29T20:13:06-07:00**.

3. Results

In this section, we investigate the two competing forces that influence the model’s final answer: the presentation order of the documents, and the semantic framing of the sources (e.g., labeling a document as an official guideline versus a recent team update). We evaluate the magnitude of these biases across different prompt formats, test whether they are driven by simple copy cues, and measure what happens when they compete directly. Note that our investigation here is strictly behavioral; mapping the specific neural circuits responsible (e.g., via activation patching) is outside the scope of this study.

3.1 Primacy is reliable in direction, highly variable in size

The following table displays the measurement of the model’s positional bias across all ten of the experiment’s test conditions:

Table 4: Positional bias across the 10 test conditions.

Cell	Header	Body	Cues	Serial Position (half- effect, nats)	95% CI	Items Show- ing Pri- macy	90% CI clears -0.15
T1	neutral	verbatim	2	-0.485	[-0.611, -0.283]	12/13	✓ [-0.592, -0.318]
T1c	neutral	verbatim	0	-0.884	[-1.074, -0.708]	13/13	✓ [-1.044, -0.735]
T2	neutral	mixed	1	-0.377	[-0.496, -0.227]	12/13	✓ [-0.478, -0.252]
T2c	neutral	mixed	0	-0.635	[-0.902, -0.452]	13/13	✓ [-0.854, -0.478]
T3	framed	verbatim	2	-0.679	[-0.848, -0.520]	13/13	✓ [-0.822, -0.543]
T3c	framed	verbatim	0	-0.569	[-0.739, -0.429]	13/13	✓ [-0.713, -0.452]
T4	framed	mixed	1	-0.172	[-0.278, -0.066]	10/13	× [-0.262, -0.082]

Cell	Header	Body	Cues	Serial Position (half- effect, nats)	95% CI	Items Show- ing Pri- macy	90% CI clears -0.15
T4c	framed	mixed	0	-0.482	[-0.603, -0.345]	13/13	✓ [-0.586, -0.366]
T5	framed	loaded	2	-0.205	[-0.308, -0.068]	11/13	× [-0.293, -0.091]
T5b	framed	loaded	0	-0.265	[-0.421, -0.137]	10/13	✓ † [-0.394, -0.159]

(Note: For detailed definitions of all table columns and metrics, please see Appendix E.)

Outcome:

- **Primacy is real and points the same way every time.** Its magnitude is a property of the prompt template, not of the model, and must never be quoted from a single template. In our testing, primacy was significant in **10/10** conditions.
- **The magnitude of primacy is highly variable.** It ranged from -0.172 to -0.884 nats, a spread by a factor of ~5 produced *entirely by surface wording* (e.g., changing ‘Agile is the system’ to ‘Agile represents our system’) while the actual disagreement stayed identical. In T4 and T5, the direction was reliable but the magnitude couldn’t be distinguished from trivially small.

3.2 Primacy is not caused by a repeated copy cue

We tested the theory that a simple copy-and-paste mechanism, triggered by repeated word patterns (e.g. “**is** [Answer]”), was causing the model to mechanically favor the first document it read. If this were true, deleting the repeated phrase while holding everything else fixed should weaken the primacy effect (which would show up mathematically as a positive contrast).

The following table displays the specific comparisons (contrasts) used to test whether removing repeated word patterns (the copy cues) reduces the model’s bias toward the first document:

Table 5: Effect of copy-cue removal on primacy (Contrasts A1–A4).

Contrast	Description	Filtered Dataset (13 items, primary)	Unfiltered Dataset (14 items)
A1	Cue removal, neutral header (T1c - T1)	-0.400 [-0.578, -0.237] *	-0.407 [-0.573, -0.251] *
A2	Cue removal, authority header (T3c - T3)	+0.110 [-0.081, +0.287]	+0.074 [-0.112, +0.257]
A3	Average of A1 and A2, descriptive only	-0.145 [-0.290, +0.010]	-0.166 [-0.306, -0.011] *
A4	Do A1 and A2 arms differ?	-0.509 [-0.687, -0.316] *	not computed ‡

‡ Evaluated on the primary filtered dataset only. Because arms A1 and A2 are analyzed individually in Section 3.2, this comparison is not computed for the unfiltered sensitivity set.

(Note: For a comprehensive index and detailed definitions of all contrasts, please see Appendix B. For definitions of general table metrics, see Appendix E.)

Outcome:

- **Short, repeated word cues do not cause the primacy effect.** Instead, what actually increases the model’s bias toward the first document is when the two competing text blocks repeat each other entirely word-for-word. Introducing wording variation between the blocks significantly reduces this bias.
- **This is a surface-level behavioral observation.** Mechanistic tests, such as internal activation patching, required to identify the specific neural circuits responsible are outside the scope of this study.
- **The outcome is supported by four distinct lines of evidence.** Each of these is independently sufficient to refute the cue hypothesis:
 1. **A1 pointed the wrong way for the cue account.** Removing the cue under a neutral header increased primacy by 0.400 nats, contradicting the expected decrease.
 2. **A2 found nothing.** Removing the cue under an authority header produced no detectable change in the effect size.
 3. **The strongest primacy in the filtered dataset had no cue at all.** The strongest primacy effect observed in the study ($T1c = -0.884$, with 13/13 items negative) occurred in a completely cue-free condition.

4. **The cue ladder wasn’t a ladder.** The magnitude of the primacy effect does not scale with the number of cues (going from $2 \rightarrow 1 \rightarrow 0$ cues yielded $-0.485 \rightarrow -0.377 \rightarrow -0.635$), effectively refuting a linear dose-response relationship.

3.3 Framing beats position by roughly 4 to 1

To measure the effects of semantic framing versus presentation order head-to-head, we analyzed our role-bound, loaded test conditions (T5 and T5b). We first measured the overall strength of each force using the cleanest, cue-free test condition (T5b).

Table 6: Baseline forces and sanity checks in the cue-free condition (T5b).

Quantity	Estimate	95% CI
Source framing (header + body together)	+1.054	[+0.907, +1.258] *
Serial position (half-effect)	-0.265	[-0.421, -0.137] *
Residual check (block-order null control)	+0.011	✓ [-0.058, +0.087]
Token (base word preference)	+0.003	✓ [-0.135, +0.165]

(Note: For detailed definitions of these quantities and general table metrics, please see Appendix E.)

The near-zero Token value (+0.003) proved that the model’s natural vocabulary bias was completely canceled out, and the near-zero Residual check (+0.011) confirmed that no hidden positional artifacts leaked into the results.

Next, we calculated the precise, item-by-item difference between the two competing forces (source framing and serial position) using statistical contrasts:

Table 7: Item-by-item comparison of source framing versus presentation order.

Contrast	Description	Filtered Dataset ($n = 13$)	Unfiltered Dataset ($n = 14$)
D1	Framing vs. position margin, cue-free (T5b)	+0.774 [+0.585, +0.966] *	+0.806 [+0.616, +0.991] **

Contrast	Description	Filtered Dataset ($n = 13$)	Unfiltered Dataset ($n = 14$)
D2	Framing vs. position margin, with cues (T5)	+0.316 [+0.188, +0.434] *	+0.317 [+0.199, +0.428] *
D3	Effect of removing cues on framing (T5b - T5)	+0.539 [+0.330, +0.843] *	+0.583 [+0.366, +0.874] *

(Note: For a comprehensive index and detailed definitions of all contrasts, please see Appendix B. For definitions of general table metrics, see Appendix E.)

Outcome:

- **Framing overpowered position.** When we measured the difference item by item in our cleanest test condition (T5b), the impact of framing outweighed the impact of position by +0.774 nats (Contrast D1). Because this cell passed all sanity checks overwhelmingly, this proved that a document’s semantic framing was vastly more influential than its reading position. The full first-versus-last framing swing is $2 \times 1.054 = 2.108$ nats (roughly 8-to-1 odds).
- **Removing cues strengthened the effect.** We also found that removing repeated copy cues from the text actually made the framing effect significantly stronger (Contrast D3). In the test with cues (T5), the framing effect was about half as strong as it was in the completely cue-free test (T5b).
- **A Note on the 4:1 Ratio:** You might notice that dividing the overall Framing average (1.054) by the Serial position average (0.265) yielded roughly 3.98, while our system logged the ratio as 3.76x. This is not a rounding error. The 3.76x ratio was calculated by comparing the absolute values of the two forces item by item. Because 3 of the 13 items actually showed a slight recency effect (a positive Serial value), they partially canceled out the negative primacy values when averaged together, which slightly shrank the final signed average. Therefore, *roughly 4 to 1* serves as a descriptive headline, but the most robust measurement of this gap remains the item-by-item difference of +0.774 nats.

3.4 Authority headers reduce bias, body wording modulates how much

To understand how different elements of a prompt work together, we tested whether adding an authoritative header affects the model’s bias differently depending on the wording of the body text. We discovered that:

- Authority headers consistently reduce bias.
- The body wording changes how much the bias is reduced.

To ensure clean results, all three of the comparisons below were conducted on text completely free of repeated copy cues:

Table 8: The interaction between authority headers and body wording variation.

Contrast	Description	Filtered Dataset ($n = 13$)	Unfiltered Dataset ($n = 14$)
C1	Effect of authority header, mixed bodies (T4c - T2c)	+0.153 [-0.011, +0.352]	+0.123 [-0.033, +0.322]
C2	Effect of authority header, verbatim bodies (T3c - T1c)	+0.315 [+0.172, +0.420] *	+0.296 [+0.166, +0.403] *
C3	The interaction (C1 - C2)	-0.162 [-0.303, -0.034] *	-0.173 [-0.301, -0.048] *

(Note: For a comprehensive index of all contrasts, see Appendix B. For definitions of general table metrics, see Appendix E.)

Outcome:

- **Authority headers consistently reduce positional bias regardless of body wording:** Adding an authoritative header reduced the model’s tendency to favor the first document, regardless of whether the body texts were reworded or identical. Although C2 is statistically significant while the confidence interval for C1 spans zero, the significant interaction contrast (C3) confirms that body phrasing modulates the header’s attenuating effect.
- **The interaction is about magnitude, not direction:** The interaction contrast (C3) proved that the body wording changed how much the authority header reduced the bias, but it didn’t flip the direction of the effect. The authority header always reduced the primacy effect, but did so to a different degree depending on the body text.

3.5 Transparency and Robustness Checks

In the interest of full transparency, the Role `null` failed in 5 of 8 testable conditions and the block-order `null` in 1 of 10. All residuals were small — the largest, 15.1% of the headline framing effect, with the rest at or under 7% — and full per-cell values are in the released log. For more information, see `o2_results_ad_series_FINAL.txt` in the source files released with this study.

Furthermore, we re-ran all of our calculations on an unfiltered dataset (14 items) to ensure our findings were not artificially created by our outlier-screening rules. Every headline finding in this paper maintained its exact same direction and statistical significance regardless of which dataset was used. Although secondary contrasts A3 and E1 (see Appendix B) shifted in significance under the unfiltered set, neither supports a primary conclusion. Ultimately, our most critical test condition (T5b) passed its sanity checks. We note one dependency: the single item removed by our outlier rule (Spec/Brief) was flagged in T5b itself, so T5b’s clean nulls are partly a consequence of that rule rather than independent of it.

4. Discussion

Position matters; framing matters more. In the cleanest test condition where the two forces compete head-to-head, source framing moves the model’s answer by +1.054 nats while reading position moves it by only -0.265. The item-by-item difference of +0.774 nats is the most defensible single number in this paper.

Position is not a stable quantity. With the underlying factual disagreement held fixed, the model’s positional bias fluctuates by at least a factor of 5 purely based on the header wording and how verbatim the two bodies of text are. Therefore, any deployment advice of the form “put the important document first—it’s worth X nats” is unsupportable: the value of X belongs to the prompt template, not to the model. We expect this to be the most practically useful takeaway for developers.

The copy-cue explanation of primacy does not survive. If a mechanical copy-and-paste mechanism drove the model’s preference for the first document, deleting the repeated phrase should weaken that bias. Instead, removing the cue under a neutral header actually increased the bias, and removing it under an authoritative header did nothing at all. Furthermore, the strongest primacy effect in the entire study occurred in a completely cue-free condition. The surface-level pattern that does actually track with the primacy effect is verbatim repetition across the text blocks.

What this licenses mechanistically: nothing yet. Our finding regarding word-for-word repetition is a regularity about text behavior, not evidence about internal attention heads or neural circuits. Mapping these circuits would require internal activation patching across text blocks, which is outside the scope of this study.

5. Limitations and scope

5.1 Methodological caveats

In our most critical test conditions (T5 and T5b), the header and the body text operate together as a single, inseparable package. While we successfully counterbalanced the specific answer words to cancel out vocabulary bias, we deliberately did not swap the loaded body text between roles. As noted in Section 2.2, counterbalancing loaded bodies across roles would have nullified the semantic manipulation by symmetry.

Because of this design, our primary effect of +1.054 nats represents the effect of the source framing as a complete package. Our current data cannot split this number to tell us how much of the effect came from the header alone versus the body text alone. Calculating that exact split would require running an additional test condition (pairing an authority header with a neutral body), which we did not perform in this study.

A second, structural aliasing affects our verbatimness comparisons. Cells in which both blocks share a template admit only one wording-to-block assignment, whereas mixed-wording cells admit two and are averaged over both. Verbatimness is therefore aliased with counterbalancing depth: the two cannot be separated, because a shared template leaves nothing to swap. Any interaction between wording and position would attenuate the mixed-wording estimates by averaging. This affects the copy-cue contrasts, the authority-header interaction, and the endpoints of the position-bias spread. Readers wishing to inspect the per-assignment values will find them in the released log.

5.2 What our results do not claim

To clarify the negative scope of our findings and prevent over-generalization, we explicitly note several conclusions that our data does not support:

- Our evaluation characterizes behavioral input–output relationships only. We do not isolate attention heads, trace residual streams, or identify circuits, and make no claims regarding internal mechanistic representations.
- We make no claims regarding whether semantic contradiction inherently suppresses positional bias; in our data, this contrast was unstable and sensitive to outlier filtering.
- Failing to reject non-equivalence within our TOST bounds (± 0.15 nats) does not prove that the underlying effects are large, only that we cannot statistically bound them as negligible.
- The observed $\approx 3.76 : 1$ ratio between semantic framing and presentation order is descriptive of condition T5b and should not be interpreted as an invariant model parameter across other prompts or architectures.
- In loaded conditions (T5/T5b), headers and bodies co-vary by design. Our

data quantifies the joint package effect (+1.054 nats) and cannot allocate relative variance to headers versus bodies independently.

- Because verbatim templates admit only a single wording assignment, verbatimness is structurally confounded with counterbalancing depth; we report the empirical association without claiming causal independence.

5.3 Scope

The scope of our study was highly controlled, meaning our findings should not be broadly extrapolated without further testing. Specifically, our experiment was limited to a single model (google/gemma-4-e4b) and evaluated using only two-document contexts, single-word answers, a single question format, and English text.

Because all of our test items were English project-management nouns, we have not verified whether these behavioral patterns generalize to other topics or vocabularies.

Finally, these findings should not be automatically applied to real-world, multi-document Retrieval-Augmented Generation (RAG) systems operating at realistic context lengths until they can be formally replicated. Our evaluation relies on short, single-turn prompts where the model processes all text with full fidelity. We do not account for the context compression, summarization, or KV-cache eviction mechanisms often utilized in long-context agent harnesses or multi-turn chats.

6. Reproducibility and provenance

- **Project files:** To review the methods and results used in this paper, you can review the project files at <https://github.com/GoogleCloudPlatform/devrel-demos/tree/main/ai-ml/gemma-priority-probing>.
- **Log of record:** The primary log is stored at `resources/o2_results_ad_series_FINAL.txt` (and `.json`), timestamped 2026-07-29T20:13:06-07:00. We have certified that this file is byte-identical to the raw archive copy stored at `internal/archive/AD_series_raw_results/o2_results_ad_series.txt`.
- **Environment:** The experiment was executed using macOS-15.7.7-arm64, Python 3.12.13, torch 2.13.0, transformers 5.14.1, numpy 2.5.1, and scipy 1.18.0. The tests were run in evaluation mode at full 32-bit precision with no random sampling, utilizing 784 forward passes across 10 conditions with a fixed random seed of 42.

Note: The code must be executed under the `venv_torch` environment, which carries the J-Lens toolkit; note that no J-Lens or Jacobian-lens component is used by `run_ad_series.py`. This is an interpreter-provenance note only.

- **Code:** The script `run_ad_series.py` is available in `github/src/` and `internal/docs/project_1_behavioral_findings/resources/`. To ensure reliability across different file versions, all citations in this paper reference function names and section letters rather than line numbers.
- **Provenance disclosure (A4):** For full transparency, the A4 arm-heterogeneity test was committed to the codebase immediately after the primary log was generated. The reported A4 value of -0.509 [-0.687, -0.316] was produced by running this logic on the cached JSON data from that exact same run. Any minor 0.001 discrepancies when manually subtracting A2 from A1 are purely the result of three-decimal display rounding, not an estimation difference.

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Appendix A — Why counterbalancing is necessary

This appendix puts a concrete number on why our counterbalanced testing design (described in Section 2.1) is strictly necessary. The pilot data discussed below did not meet our publication standards, and it does not support our main claims. However, we include it to illustrate the massive bias that occurs without counterbalancing.

- **A.1 The cost of vocabulary bias:**

When we ran tests without counterbalancing, the model’s natural preference for certain words created a massive 1.58-nat artificial swing in the results. In a standard test, this vocabulary bias is completely indistinguishable from a real positional effect. To put this in perspective, this artificial 1.58-nat swing is 6 times larger than the positional bias half effect we measured in our clean test. Our four-way testing design exists specifically to cancel this out.

- **A.2 The pilot patching cells:**

We also generated a set of pilot data (patching cells) that we ultimately deemed unusable for aggregate analysis because too many of the test items failed our strict quality checks.

- **A.3 Why this is an appendix:**

Neither of these pilot artifacts meets the standard of the main text. We report them here purely for transparency, to prove the exact size of the artifact we successfully removed from our primary data.

Appendix B — Complete contrast index

This appendix provides a comprehensive index of all the comparisons (contrasts) calculated throughout the study. It includes the results for both our primary Filtered Dataset ($n = 13$) and our unfiltered dataset ($n = 14$). All reported values represent half-effects measured in nats. An asterisk (*) indicates a statistically significant result, meaning the 95% confidence interval excludes zero.

Table 9: Comprehensive index of experimental contrasts.

ID	Purpose	Filtered Dataset ($n = 13$)	Unfiltered Dataset ($n = 14$)	Status
A1	Neutral cue removal	-0.400 *	-0.407 *	Inferential
A2	Authority cue removal	+0.110	+0.074	Inferential (null)
A3	Pooled cue removal	-0.145	-0.166 *	Descriptive only
A4	Arm heterogeneity test	-0.509 *	not computed	Inferential; post-dates log
B1	Partial cue removal	+0.108	+0.110	Descriptive (confounded)
B2	Mixed-body cue removal	-0.258 *	-0.275 *	Descriptive (confounded)
B3	Total cue removal (confounded)	-0.150	-0.165	Descriptive (confounded)
B4	Verbatimness at zero cues	+0.250 *	+0.243 *	Inferential
C1	Authority header effect (mixed body)	+0.153	+0.123	Inferential (null)
C2	Authority header effect (verbatim body)	+0.315 *	+0.296 *	Inferential
C3	Cue-matched interaction	-0.162 *	-0.173 *	Headline
C4	Authority header effect (cued verbatim)	-0.194	-0.186	Legacy, cue-confounded

ID	Purpose	Filtered Dataset ($n = 13$)	Unfiltered Dataset ($n = 14$)	Status
C5	Cue- confounded interaction	+0.347 *	+0.308 *	Legacy, superseded by C3
C6	Authority header effect (cued mixed)	+0.205 *	+0.198 *	Legacy, cue- confounded
D1	Framing-vs- position margin	+0.774 *	+0.806 *	Headline
D2	Framing-vs- position margin (cued)	+0.316 *	+0.317 *	Inferential
D3	Effect of cue removal on source framing	+0.539 *	+0.583 *	Inferential
E1	Semantic suppression test	+0.216	+0.237 *	Policy- dependent
E2	Semantic suppression test (cued / confounded)	-0.033	-0.038	Confounded, reference only

Definitions:

The following section provides plain-English definitions for the specific comparisons (contrasts) indexed in the table above. Each entry explains what the calculation measures and how it isolates the different variables influencing the model’s decisions.

- **A1 (T1c - T1):** Neutral cue removal. The effect of deleting the copy cue under a neutral header, keeping the wording verbatim.
- **A2 (T3c - T3):** Authority cue removal. The same deletion, but tested under an authority header.
- **A3 (mean(A1, A2)):** Pooled cue removal. The plain average of A1 and A2. This is invalid for testing because A4 shows the two arms react too differently to be safely averaged, so it is retained for reference only.
- **A4 (A1 - A2):** Arm heterogeneity test. Tests whether the model responds to cue removal the same way under both headers. It does not,

which triggers the pooling warning.

- **B1 (T2 - T1):** Partial cue removal. Stepping one step down the cue ladder from 2 cues to 1. This is confounded because it also changes wording variation at the same time.
- **B2 (T2c - T2):** Mixed-body cue removal. Isolates the exact effect of removing the final copy-cue when the text is already mixed.
- **B3 (T2c - T1):** Total cue removal (confounded). Jumps straight from 2 cues (T1) down to 0 cues (T2c). It is “confounded” because it removes the cues and introduces mixed wording at the exact same time.
- **B4 (T2c - T1c):** Verbatimness at zero cues. The clean test to determine whether word-for-word repetition across blocks is what drives primacy.
- **C1 (T4c - T2c):** Authority header effect (mixed body). Both cells are cue-free and have mixed text. Isolates the pure effect of swapping a neutral header for an authority header.
- **C2 (T3c - T1c):** Authority header effect (verbatim body). Both cells are cue-free and have “verbatim” (identical) text. It isolates the pure effect of the authority header under strict verbatim conditions.
- **C3 (C1 - C2):** Cue-matched interaction. Checks whether the authority header’s effect on primacy depends on the body wording, holding the cue count at zero.
- **C4 (T3 - T1):** Authority header effect (cued verbatim). This is a legacy contrast. It measures the authority header effect, but it does so while the “is {e}” copy-cue is still present in the text.
- **C5 (C1 - C4):** Cue-confounded interaction. An old, flawed interaction test. It compared a cue-free arm (C1) against a cued arm (C4). Because it accidentally tested header framing and cue removal simultaneously, it was superseded by C3 (the clean, cue-matched interaction).
- **C6 (T4 - T2):** Authority header effect (cued mixed). A legacy contrast. It measures the authority header effect, but with 1 cue present and mixed text.
- **D1 (|Role| - |Serial|, T5b):** Framing-vs-position margin. The per-item, interval-bearing measurement version of the “4:1” headline.
- **D2 (|Role| - |Serial|, T5):** Framing-vs-position margin (cued). This compares the strength of semantic framing against presentation order, but does so in the cued condition (T5). It mirrors our headline D1 finding, just with the copy-cue left in the text.
- **D3 (T5b - T5 Role):** Effect of cue removal on source framing. Isolates how much the model’s reliance on semantic framing changes when you delete the copy-cue. (Since $D3 = +0.539$ nats, it proves that removing

the mindless copy-cue actually forces the model to rely more heavily on the semantic authority headers).

- **E1 (T5b - T4c):** Semantic suppression test. Tests whether a genuine clash in meaning between documents damps primacy (currently unresolved).
 - **E2 (T5 - T4):** Semantic suppression test (cued / confounded). This contrast is confounded and is retained for reference only. Originally it was meant to measure high-level semantic suppression.
-

Appendix C — The four crossings of block order x entity-role assignment

This appendix illustrates the core experimental design used to mathematically isolate the model’s biases. By testing each candidate word pair across four specific configurations, we counterbalance both the presentation order of the documents (which block is read first) and the roles assigned to each word (which word acts as the authoritative or recent source). The table below demonstrates these four crossings using the candidate pair Agile/Waterfall as an example.

Table 10: Four-way crossing configurations.

Crossing	Block order	Entity-role assignment	Resulting prompt structure
1	Authority first	Agile = Stale, Waterfall = Recent	Official Guideline: Use Agile. Team Update: Use Waterfall.
2	Authority first	Waterfall = Stale, Agile = Recent	Official Guideline: Use Waterfall. Team Update: Use Agile.
3	Recent first	Agile = Stale, Waterfall = Recent	Team Update: Use Waterfall. Official Guideline: Use Agile.
4	Recent first	Waterfall = Stale, Agile = Recent	Team Update: Use Agile. Official Guideline: Use Waterfall.

Definitions:

- **Crossing:** The specific configuration of the prompt being tested. There are four total crossings to fully swap and isolate every variable.
- **Block order:** The sequence in which the text blocks are presented to the model (i.e., whether the authoritative document or the recent update is read first). This allows us to measure positional bias.
- **Entity-role assignment:** Which specific answer word (the entity, e.g., Agile or Waterfall) is paired with which semantic framing label (the role, e.g., the stale guideline or the recent update). Swapping this allows us to cancel out the model’s natural vocabulary bias.
- **Resulting prompt structure:** A conceptual illustration of what the final text looks like when the selected block order and entity-role assignment are combined.

Appendix D — Reading nats, and cued vs. cue-free prompts

This appendix contains two reference guides for interpreting the study’s results. The first table translates nats (our unit of measurement for model bias) into standard odds and percentages. The second table provides exact examples of the cued versus cue-free prompt wording used to test the model’s reliance on repeated phrasing.

Nats

Table 11: Conversion of log-odds (nats) to probabilities and relative odds.

Nat value	CI	Odds (A to B)	Prob. of A	Prob. of B	Model’s preference
+2.0	[+1.80, +2.20]	7.39 to 1	88%	12%	Overwhelming preference for Option A.
+1.0	[+0.80, +1.20]	2.72 to 1	73%	27%	Strong preference for Option A.
0	[-0.20, +0.20]	1.00 to 1	50%	50%	Indifference. The model is guessing.
-1.0	[-1.20, -0.80]	1 to 2.72	27%	73%	Strong preference for Option B.
-2.0	[-2.20, -1.80]	1 to 7.39	12%	88%	Overwhelming preference for Option B.

Definitions:

- **Nat value:** The measured effect size in nats (log-probability units). It quantifies exactly how strongly the model is pulling toward one answer (Option A) over the other (Option B).
- **CI:** The 95% Confidence Interval. It shows the expected range of the nat value, confirming that the measurement is statistically reliable.
- **Odds (A to B):** The mathematical translation of the nat value into a standard odds ratio, showing how many times more likely the model is to choose Option A compared to Option B.
- **Prob. of A:** The implied probability (as a percentage) that the model will choose Option A, derived directly from the nat value.
- **Prob. of B:** The implied probability (as a percentage) that the model will choose Option B.
- **Model’s preference:** A plain-English description of what these numbers actually mean for the model’s behavior (e.g., whether the AI is highly opinionated or just guessing).

Cue vs Cue-Free Prompts

Table 12: Example prompt templates for cued and cue-free conditions.

Type	Role	Wording
Cued	Authority	Official guidelines state that for project planning, the required configuration is {e}.
Cue-Free	Authority	{e} is the required configuration for project planning, per official guidelines.
Cued	Recency	Update: The new updated specification for project planning is now {e}.
Cue-Free	Recency	{e} is the newly updated specification for project planning, effective now.

Definitions:

- **Type:** Indicates whether the prompt includes a repeated word pattern (Cued) or if that mechanical trigger has been completely removed (Cue-Free).
- **Role:** The semantic framing applied to the text block, presenting it either as an authoritative document (Authority) or a recent update (Recency).
- **Wording:** The exact template string fed to the model during the experiment. The {e} represents where the specific answer word (the entity) is dynamically injected into the text.

Appendix E — Results table definitions

3.1 Positional bias across 10 test conditions

The following table displays the measurement of the model’s positional bias across all ten of the experiment’s test conditions:

Cell	Header	Body	Cues	Serial Position (half- effect, nats)	95% CI	Items Show- ing Pri- macy	90% CI clears -0.15
T1	neutral	verbatim	2	-0.485	[-0.611, -0.283]	12/13	✓ [-0.592, -0.318]
T1c	neutral	verbatim	0	-0.884	[-1.074, -0.708]	13/13	✓ [-1.044, -0.735]
T2	neutral	mixed	1	-0.377	[-0.496, -0.227]	12/13	✓ [-0.478, -0.252]
T2c	neutral	mixed	0	-0.635	[-0.902, -0.452]	13/13	✓ [-0.854, -0.478]
T3	framed	verbatim	2	-0.679	[-0.848, -0.520]	13/13	✓ [-0.822, -0.543]
T3c	framed	verbatim	0	-0.569	[-0.739, -0.429]	13/13	✓ [-0.713, -0.452]
T4	framed	mixed	1	-0.172	[-0.278, -0.066]	10/13	× [-0.262, -0.082]
T4c	framed	mixed	0	-0.482	[-0.603, -0.345]	13/13	✓ [-0.586, -0.366]
T5	framed	loaded	2	-0.205	[-0.308, -0.068]	11/13	× [-0.293, -0.091]
T5b	framed	loaded	0	-0.265	[-0.421, -0.137]	10/13	✓ † [-0.394, -0.159]

Definitions:

- **Cell:** The specific test condition being evaluated. The possible results are

the ten conditions created for the study: T1, T1c, T2, T2c, T3, T3c, T4, T4c, T5, and T5b.

Suffixes on the test conditions (such as the ‘c’ in T1c or the ‘b’ in T5b) denote the cue-free (0-cue) variants of the base tests. These specific identifiers are retained to match the exact condition names logged in the open-source repository.

- **Header:** The wording style used for the labels on the text blocks. The possible results are *neutral* (labeled blandly with no bias), or one of these framed types: *authority* (official guideline) or *recency* (new update). Which one is printed first is counterbalanced.
- **Body:** The wording style used within the actual text blocks. The possible results are: *verbatim* (word-for-word identical), *mixed* (reworded), or *loaded* (carrying the manipulation directly in the sentence).
- **Cues:** The number of times a specific, exact word pattern appears in the text, which researchers tested to see if it triggered a copy-and-paste mechanism in the model. The possible results are 0, 1, or 2.
- **Serial Position (half-effect, nats):** This number measures the model’s positional bias in *nats* (a unit of information). A negative number indicates a *primacy effect* (the model favors the first block it reads), while a positive number indicates a *recency effect* (the model favors the last block).
- **95% CI:** The 95% confidence interval, showing the spread of results generated from resampling the test items 9,999 times to ensure the effect is reliable. The results are shown as a range.
- **Items Showing Primacy:** How many of the 13 word pairs (items) tested actually exhibited a negative serial number (a primacy effect).
- **90% CI clears -0.15:** Answers whether the effect is large enough to be considered non-trivial, specifically checking if the 90% confidence interval lies entirely beyond a -0.15 boundary. The results are either a checkmark (✓) along with the interval if it clears the boundary, or a cross (×) if it fails to clear it, with one result receiving a dagger (†) for passing by a tiny margin.

Group definitions:

- **Cell, Header, Body, and Cues:** The test conditions. These outline the specific combination wording and structural settings used to create a particular test.
- **Serial Position:** Shows how much the model leaned toward a document based solely on where it was placed in the prompt. A negative number indicates a primacy effect while a positive number indicates a recency effect.

- **95% CI, Items Showing Primacy, and 90% CI clears -0.15:** Statistical proof to confirm that the measured effect was consistent across the tested items and large enough to be considered non-trivial.

3.2 Contrasts

The following table displays the specific comparisons (contrasts) used to test whether removing repeated word patterns (the copy cues) reduces the model’s bias toward the first document:

Contrast	Description	Filtered Dataset (13 items, primary)	Unfiltered Dataset (14 items)
A1	Cue removal, neutral header (T1c - T1)	-0.400 [-0.578, -0.237] *	-0.407 [-0.573, -0.251] *
A2	Cue removal, authority header (T3c - T3)	+0.110 [-0.081, +0.287]	+0.074 [-0.112, +0.257]
A3	Average of A1 and A2, descriptive only	-0.145 [-0.290, +0.010]	-0.166 [-0.306, -0.011] *
A4	Do A1 and A2 arms differ?	-0.509 [-0.687, -0.316] *	not computed ‡

Definitions:

- **Contrast:** Each row represents a different test measuring what happens when the copy cue is deleted. Each test is defined as an arm.
 - **A1:** Shows the effect of deleting the cue when the documents have neutral, unbiased titles.
 - **A2:** Tests deleting the cue when the documents have official, authoritative titles.
 - **A3:** Calculates the average between the first two tests (A1, A2).

We did not average the two arms A1 and A2. Statistically, two measurements should only be averaged if they capture the same underlying effect. Our analysis script automatically tested this by calculating the difference between the two conditions. The resulting confidence interval excluded zero, proving the two arms differ significantly. Consequently, the script issued a warning against pooling these heterogeneous results, so we reported A1 and A2 separately and treated their average (A3) purely as a descriptive metric with zero inferential weight.

- **A4:** Calculates the difference between the first two tests (A1, A2).
 - ‡ Evaluated on the primary filtered dataset only. Because arms A1 and A2 are analyzed individually in Section 3.2, this comparison is not computed for the unfiltered sensitivity set.
- **Filtered Dataset:** The results using the strict, primary dataset of 13 word pairs that passed all sanity checks. The result is presented in this format: *contrast [95% confidence interval] **
- **Unfiltered Dataset:** A sensitivity check where the researchers re-ran the calculations with 14 word pairs (including one they had previously excluded) to make sure the overall conclusions didn’t change. The result is presented in this format: *contrast [95% confidence interval] **
- **contrast [95% confidence interval] *:**
 - **contrast:** The difference between two different test conditions. A negative number represents a primacy effect, a positive number represents a recency effect.
 - **95% confidence interval:** Shows the spread of results generated by resampling 9,999 times to ensure the measured effect was reliable.
 - **(*):** The 95% confidence interval does not include zero. When the interval excludes zero, it means the measured effect is statistically significant and consistently points in the same direction across all the resampled test variations.

3.3 Framing and position influence

The following table contains the test conditions we used to measure framing and position influence:

Test Condition	Header Wording	Body Wording	Cue Count
T5	framed	loaded	2
T5b	framed	loaded	0

Definitions:

- **Test Condition:** The specific experimental setup being evaluated. The body text acts as the manipulation and is role-bound to the header.
 - **T5:** Repeated word patterns (copy cues) are present.
 - **T5b:** All repeated word patterns (copy cues) are completely removed.

- **Header Wording:** The wording style used for the labels on the text blocks. In these specific tests, they are *framed* (present the document as an official guideline or a fresh update).
- **Body Wording:** The wording style used within the actual text blocks. In these tests, the body is loaded (the authoritative or recent tone is embedded directly within the sentences themselves).
- **Cue Count:** The number of times a specific, exact word pattern (e.g. “is [Answer]”) appears in the text. We used this to test if the model was relying on a simple copy-and-paste mechanism.

3.3 Quantities for hidden artifacts

The following table contains the quantities we measured for hidden artifacts and natural vocabulary biases in T5b:

Quantity	Estimate	95% CI
Source framing (header + body together)	+1.054	[+0.907, +1.258] *
Serial position (half-effect)	-0.265	[-0.421, -0.137] *
Residual check (block-order null control)	+0.011	✓ [-0.058, +0.087]
Token (base word preference)	+0.003	✓ [-0.135, +0.165]

Definitions:

- **Quantity:** The specific variable, force, or sanity check that we measured in test condition T5b.
 - **Source framing:** How much the authoritative or recent tone of the document influenced the model’s answer.
 - **Serial position:** How much the order of the documents (first versus last) influenced the model’s answer.
 - **Residual check:** A sanity check. The near-zero Order value (+0.011) confirms no hidden positional artifacts leaked into the results.
 - **Token:** A sanity check. The near-zero Token value (+0.003) proves that the model’s natural vocabulary bias was completely canceled out.
- **Estimate:** The calculated measurement of the effect, expressed in nats. It shows how hard the model’s decision was pulled in a certain direction.

- **95% CI:** The 95% confidence interval. This shows the spread of results generated by resampling the test items 9,999 times to ensure the measurement is reliable. An asterisk (*) means the effect is statistically significant because the range does not include zero.

3.3 Influential factors

We use the following contrasts to separate the different factors that influence the model’s decisions:

Contrast	Description	Filtered Dataset ($n = 13$)	Unfiltered Dataset ($n = 14$)
D1	Framing vs. position margin, cue-free (T5b)	+0.774 [+0.585, +0.966] *	+0.806 [+0.616, +0.991] **
D2	Framing vs. position margin, with cues (T5)	+0.316 [+0.188, +0.434] *	+0.317 [+0.199, +0.428] *
D3	Effect of removing cues on framing (T5b - T5)	+0.539 [+0.330, +0.843] *	+0.583 [+0.366, +0.874] *

Definitions:

- **Contrast:** A calculation that measures the exact difference between test conditions to separate the various factors influencing the model’s decisions.
 - **D1:** Measures the head-to-head difference between the framing effect and the positional effect in the completely cue-free condition (T5b).
 - **D2:** Measures that exact same head-to-head comparison, but in the test condition that included copy cues (T5).
 - **D3:** Calculates the difference in the framing effect between the cued test and the cue-free test.
- **Filtered Dataset:** The results using the strict, primary dataset of 13 word pairs that passed all sanity checks. The result is presented in this format: *contrast [95% confidence interval] **
- **Unfiltered Dataset:** A sensitivity check where the researchers re-ran the calculations with 14 word pairs (including one they had previously excluded) to make sure the overall conclusions didn’t change. The result is presented in this format: *contrast [95% confidence interval] **

Note: In all three contrasts, the asterisk (*) confirms that the 95% confidence intervals do not include zero, meaning all of these findings are statistically significant.

3.4 Authority headers

To ensure clean results, all three of the comparisons below were conducted on text completely free of repeated copy cues:

Contrast	Description	Filtered Dataset ($n = 13$)	Unfiltered Dataset ($n = 14$)
C1	Effect of authority header, mixed bodies (T4c - T2c)	+0.153 [-0.011, +0.352]	+0.123 [-0.033, +0.322]
C2	Effect of authority header, verbatim bodies (T3c - T1c)	+0.315 [+0.172, +0.420] *	+0.296 [+0.166, +0.403] *
C3	The interaction (C1 - C2)	-0.162 [-0.303, -0.034] *	-0.173 [-0.301, -0.048] *

Definitions:

- **Contrast:** A calculation used to measure the effect of an authoritative header and test whether that effect changes depending on the wording of the body text.
 - **C1:** Measures how much an authoritative header changes the model’s bias when the competing documents are reworded (mixed).
 - **C2:** Measures how much an authoritative header changes the model’s bias when the competing documents are word-for-word identical (verbatim).
 - **C3:** Calculates the exact difference between C1 and C2 to test whether the body wording significantly alters the header’s impact.
- **Filtered Dataset:** The results using the strict, primary dataset of 13 word pairs that passed all sanity checks. The result is presented in this format: *contrast [95% confidence interval] **

Glossary

- **Activation patching:** A mechanistic interpretability technique used to identify specific neural circuits and attention heads by replacing intermediate hidden activations during inference.
- **AD series:** The final experimental iteration (recorded in `AD_series_FINAL.txt`) that resolved design limitations of earlier exploratory series (AA through AC) and provides the definitive dataset for this paper.
- **Behavioral suite:** An evaluation framework measuring model input–output behavior under controlled prompt variations from the outside, without assuming internal representations.
- **Block order:** The presentation order in which competing document blocks are delivered to the model (e.g., whether Block 1 or Block 2 appears first).
- **CI:** Confidence Interval. See *Confidence Interval*.
- **Confidence Interval:** A statistical range (here, 95% BCa bootstrap with 9,999 resamples) estimating the plausible spread of an effect size across test items.
- **Copy cue:** A specific word pattern suspected of triggering mechanical copying. See *Cue*.
- **Counterbalancing:** Systematically permuting variables (document order and entity-role assignments) across configurations so baseline biases cancel out by symmetry.
- **Cue:** A repeated surface word pattern (e.g., “*is {e}*”, “*is now {e}*”) hypothesized to mechanically bias model predictions toward an answer.
- **Entity:** A specific candidate answer noun (e.g., *Scrum*, *Kanban*, *Agile*, *Waterfall*).
- **Entity-role assignment:** The mapping determining which candidate entity plays which semantic role in a prompt (e.g., assigning *Agile* to the authoritative guideline and *Waterfall* to the team update, or vice versa).
- **Forward pass:** A single evaluation run feeding an input prompt through the model to compute next-token output logits.
- **Four crossings:** The 2×2 factorial design crossing block orders with entity-role assignments to isolate true behavioral effects while canceling out positional and lexical biases (Appendix C).
- **Lexical prior:** See *Vocabulary bias*.
- **Mechanistic claim:** An assertion attributing a model’s behavior to specific internal computational structures (e.g., attention heads or circuits), distinct from purely behavioral observations of input–output relationships.
- **Nat:** The standard unit of information based on natural logarithms (base e), where $1 \text{ nat} \approx 1.44 \text{ bits}$. Used to quantify log-odds shifts and log-probability differences in model predictions.
- **Positional bias:** Systematic preference for an answer based purely on its position in a prompt (e.g., primacy or recency), independent of content.
- **Primacy:** Systematic bias favoring information presented earlier in a prompt.

- **Recency:** Systematic bias favoring information presented later in a prompt.
- **Serial position:** The presentation order of documents in context; one of the two primary forces measured in this study.
- **Source framing:** The semantic framing or authority attributed to a document (e.g., an official guideline versus a team update); the dominant force measured in this study.
- **Statistical model:** A mathematical model that assigns probabilities to token sequences based on patterns learned during pretraining.
- **Surface wording:** The superficial phrasing used to present information without altering the underlying factual proposition (e.g., verbatim repetition vs. paraphrased body text).
- **Vocabulary bias (lexical prior):** The model’s baseline preference for one word over another regardless of context (e.g., naturally preferring *Agile* over *Waterfall*).